**4** Fig. 4.1 shows an RC circuit consisting of a cell of electromotive force 6.0 V, two capacitors of capacitances $2.0 \times 10^{-3}$ F and $3.0 \times 10^{-3}$ F, and an ohmic resistor of resistance 400 Ω. Both capacitors are initially uncharged. The switch is then closed to allow the capacitors to charge.

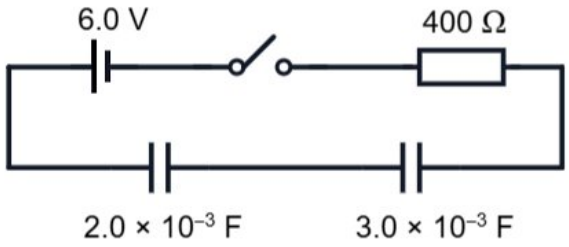



**Fig. 4.1**

**(a)** Explain why, at any instant during the charging process, the charge stored on the two capacitors is equal.

.....

..... [1]

**(b)** Hence, show that the potential difference across the $3.0 \times 10^{-3}$ F capacitor is 2.4 V when it is fully charged.

[2]

**(c)** Hence, show that the charge stored by the $3.0 \times 10^{-3}$ F capacitor when it is fully charged is $7.2 \times 10^{-3}$ C.

[1]

**(d)** Determine the time taken for the $3.0 \times 10^{-3}$ F capacitor to charge from being uncharged to storing $4.0 \times 10^{-3}$ C of charge.

time = ..... s [2]

**(e)** Calculate the change in the potential energy stored in the $3.0 \times 10^{-3}$ F capacitor as its charge increases from $2.5 \times 10^{-3}$ C to $4.5 \times 10^{-3}$ C.

energy = ..... J [2]

**(f)** Explain why the energy required to transfer each successive unit of charge to a capacitor increases as it is being charged.

.....

.....

.....

..... [2]